

Painter Names and Feature Distributions: Comparing Generated Images with Original Paintings

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Abstract

An image may evoke a painter while occupying a different distribution of measurable properties. We compare 1,006 newly generated images from Nano Banana 2, FLUX.2 Max and a GPT Image 2 service with 70 fresh digital painting surrogates by Monet and Cézanne. A fixed 24-brief design includes artist-free and named controls, three repetitions and an additional generic/detailed intervention. Thirty-one interpretable features are scaled using separate historical development works; content weighting and processing sensitivities accompany energy discrepancy, spread, coverage and grouped detection. In the primary comparison, the eight named collections have generated/original variance ratios .306–.756, yet their full-feature RBF balanced accuracies remain .893–.987. Naming reduces energy discrepancy in all six route/painter point estimates; the four paid-route contrasts reject the prespecified conditional randomization null after Holm correction, whereas the OAuth naming and detail contrasts do not. Paid artist-free variance ratios are 1.069–1.535, so concentration is not a universal property of every generated condition. Paid-route improvements persist descriptively under both processing sensitivities, while reference restriction and feature families show exceptions to uniform contraction. These finite-panel findings distinguish improved proximity from recovery of original variation. They do not establish perceptual style equivalence; reference capture and variable service rendering remain unresolved. All collection, measurement and numerical replay checks are complete, with provider-reported charges of \$40.68.

1 Introduction

The question motivating this study is whether painter-conditioned image generation reproduces the distribution of visual properties found across a painter’s original works. A small distance to a painter’s average feature vector does not answer this question: generated images can cluster tightly near that average while failing to reproduce variation across works. Conversely, a recognizable artistic reference may coexist with systematic differences in palette, spatial organization or digital texture.

We distinguish four properties: distribution discrepancy, within-set spread, reference coverage and painter specificity. None is an aesthetic quality score. The measured objects are digital reproductions of paintings and generated image files. Their comparison combines artistic choices with content, reproduction, resizing, encoding and service behavior. Treating every detectable difference as a failure of painter-style reproduction would overstate what the measurements identify.

The completed analysis suggests a useful, narrower hypothesis: under the tested prompt libraries and image processing, painter-conditioned generations occupy more concentrated and distinguishable feature distributions than the recorded original-painting surrogates. The controlled study tests whether this pattern persists under stronger content controls and across

additional model families. Its central finding is that painter naming can improve proximity while contracting measured spread; artist-free controls show why these properties should not be conflated.

2 Related work and the distinct research question

Quantitative art research has characterized painting collections using color, image statistics and information-theoretic measures (Kim et al., 2014; Lee et al., 2020; Sigaki et al., 2018). Kim et al.’s 2026 study concerns contextual multimodal navigation through art-historical development (Kim et al., 2026). Our sampling unit and target differ: we compare within-painter distributions across distinct works with newly generated text-to-image collections, with painter-name and prompt-content interventions. We do not claim to replicate its art-evolution analysis or its complete measurement system. Our fixed 31-feature representation is the existing implementation in this project.

Asperti’s recent preprint reports separation of human and generated paintings in CLIP space and investigates preprocessing robustness, interpretable multiscale structure and inversion (Asperti, 2026). Thus, scatter separation alone is not sufficient novelty for this work. We emphasize painter-conditional comparisons, explicit artist-free controls and the effects of prompt content on a fixed interpretable representation. We have not run CLIP or reproduced that preprint’s inversion experiments. More generally, feature-space distinctions need not have the perceptual significance assumed by a style evaluator.

Generative-model evaluation separates fidelity and coverage for related reasons (Naeem et al., 2020). Resizing and compression can change measured differences (Parmar et al., 2022); digital-surrogate provenance is consequently part of the experimental design rather than a minor file-handling detail. Energy statistics provide a distributional comparison based on pairwise distances (Székely and Rizzo, 2013), without reducing each collection to its centroid.

3 Development evidence and design motivation

The development analysis compares 649 digital painting surrogates (297 Monet, 106 Sisley, 141 Pissarro and 105 Cézanne) with 1,920 generated images. Its grid crosses two requested GPT Image service aliases, three prompt methods, 16 scene templates and four repetitions, with named and artist-free conditions. Two later, explicitly authorized exact-payload retries complete the descriptive grid after its original two refusals: 1,920 images from 1,922 requests. Their later timing does not restore the original complete-grid primary. These development findings are descriptive; they have no retrospective replacement-based inference. The requested aliases do not attest distinct underlying model snapshots.

Named cells have generated/original sample-variance ratios .206–.376 and scene/work-disjoint RBF balanced accuracy .940–.983. Artist-free cells also separate strongly (.915–.990), with larger spread ratios (.369–.750). Reweighting coarse title/prompt-derived content classes changes the variance ranges to .211–.372 and .378–.728, respectively. Matched-size disjoint original splits have much smaller median energy discrepancies (.1635–.2149) than named/original comparisons (1.6776–2.1419). These repeated splits describe one finite corpus, not independent studies or a calibrated equivalence margin.

Earlier family-specific distance comparisons favor by-name prompting in 19 of 24 combinations of alias, painter and feature family; style-plus-aspects is closest in five. Explicit style instructions increase distance relative to by-name prompting in all 24; adding aspects has mixed directions. Cross-alias RBF transfer remains strong (median balanced accuracy .965), consistent with shared generated-image or capture signatures. Together, these observations motivate model-family

breadth, better content labels and a focused prompt intervention. They do not establish best achievable prompting or isolate a painter-style mechanism. The completed historical report retains every numeric result, projection and split membership; its different reference corpus is not pooled with the new study.

4 Controlled study design

4.1 Choice of expansion axis

We expand model-family breadth and improve the comparison target before adding more painters. New collection focuses on Monet and Cézanne. The existing four painter results are retained as development evidence. This is a disclosed two-painter case study, chosen after earlier outcomes were known; it cannot estimate the prevalence of the phenomenon across all painters or traditions.

The research grid uses the same 24 content briefs across both painters, with three repetitions, named and artist-free conditions on three routes, plus a generic named condition on OAuth: 1,008 research slots. Eighteen separate technical requests and at most 24 bounded transient-error retries give a ceiling of 1,050 attempts, including at most 612 paid attempts. The spending ceiling is \$75, including the pilot and retries. Successful images are never regenerated or selected among alternatives. Sample counts reflect affordability and design coverage; they do not establish power for an image-space effect.

The reference target initially specified 48 fresh works per painter and 12 content strata. The recorded identity audit selects 64 Monet and 48 Cézanne candidates outside the prior exposure indexes. Original delivery encounters Wikimedia rate limits; a separately frozen successor follows its supported thumbnail policy. Nonupsampled standard delivery supports 38 Monet and 33 Cézanne candidates. All 71 deliveries succeed. Single-maintainer LLM visual coding excludes one figure-led Cézanne, leaving 38 Monet and 32 Cézanne works. These unequal counts are disclosed before measurement; no 48-work panel or fully balanced 12-stratum design is claimed.

Visually coded water, built and land organization replaces geographic-name heuristics. Counts are 21/4/13 for Monet and 3/11/18 for Cézanne. The primary comparison gives original works equal weight and generated images the same fixed class masses as their target reference. Eight shared briefs per class support this weighting. Collected-mixture and equal-class sensitivities are also required; the latter heavily weights Cézanne’s three water scenes. Mixed annotations remain flagged, with an exclusion sensitivity. This is coarse content control, not exact scene matching or independent expert annotation. Independent photographic capture pairs have not been established. Thumbnail and alternate-file provenance cannot substitute for evidence of distinct capture events.

4.2 Technical pilot results

All six requests per route return decodable images. Google `gemini-3.1-flash-image` through the pinned Google AI Studio provider returns 1024-square JPEGs, costing \$0.4105 in total. BFL `flux.2-max` through its pinned provider returns 1024-square PNGs, costing \$0.4200. The local OAuth `gpt-image-2` comparator returns nonsquare PNGs with short sides 1121–1126 and reports low rather than requested medium quality. No paid fallback, retry, prompt rewrite or feature-based pilot selection occurs.

The completed research collection retains 1,006 images from 1,008 fixed slots: 288 Nano Banana 2, 288 FLUX.2 Max and 430 OAuth outputs. Two OAuth generic-named Cézanne slots receive moderation refusals and remain missing. One complete FLUX 502 receives one successful exact-payload technical retry under the frozen rule. Including 18 pilot attempts

and this retry, there are 1,027 physical attempts, 589 paid. Provider-reported charges total \$40.6819185. The failed FLUX call omits cost; its raw null remains, with a full \$5 contingency, giving conservative accounting \$45.6819185 against the \$75 ceiling. No unresolved outcome or unclassified cost remains. OpenRouter’s published failed-image billing waiver ([OpenRouter, 2026](#)) is not substituted for an individual receipt.

All paid research outputs are 1024-square: Google JPEG and BFL PNG. OAuth returns 430 PNGs with short sides 1087–1254 pixels; eight are square. Of these, 424 report low quality and six report medium despite all requesting medium. The six medium outputs are artist-free (three per painter); all named outputs report low. These response fields do not attest an immutable model or comparable rendering. OAuth is therefore a separate service comparison. A prompt contrast on that route includes any prompt-induced service handling; it cannot isolate painter naming at fixed delivered quality. No output is filtered by its reported quality. Symmetric 256-pixel and common-JPEG sensitivities probe processing effects but cannot undo earlier encoding or generation-quality differences.

5 Distribution measurements and statistical analysis

5.1 Feature representation and processing

Each image is represented by 11 color, eight spatial and 12 digital-texture features. Color includes lightness and chroma location/spread, hue organization, and multiscale chromatic differences. Spatial features include spectral shape, orientation organization, gradients and quadrant differences. Texture includes wavelet energies and trends, local binary-pattern entropies and local luminance variation. These are measurements of digital images, not direct measurements of physical brushstroke relief or a validated complete definition of painter style.

The primary normalization applies EXIF orientation, converts valid embedded profiles to sRGB, assumes sRGB when compatible unprofiled data are present, rejects nonopaque images and preserves aspect ratio at a 512-pixel short side. It uses Lanczos resampling without upsampling or discretionary cropping. Each coordinate is centered by the development median and divided by the development interquartile range. Development painters receive equal total mass; no generated or new reference feature is used to fit that scaler.

5.2 Distribution discrepancy and spread

Let x_i and y_j be standardized reference and generated vectors, with nonnegative weights a_i and b_j that each sum to one. We report the empirical V-statistic energy discrepancy

$$\begin{aligned}\widehat{\mathcal{E}}(X, Y) = & 2 \sum_{i,j} a_i b_j \|x_i - y_j\|_2 - \sum_{i,k} a_i a_k \|x_i - x_k\|_2 \\ & - \sum_{j,l} b_j b_l \|y_j - y_l\|_2.\end{aligned}\tag{1}$$

All pairwise terms contribute; the reference mean is not the sole comparison target. The finite-sample V-statistic has a nonzero sampling baseline even when independent samples come from one common population.

We report the ratio of total within-group sample variances for ordinary unweighted data. Weighted analyses use the population-form trace $\sum_i a_i \|x_i - \sum_k a_k x_k\|^2$ in both numerator and denominator. The sample and population definitions have different finite-sample denominators. A complementary robust summary divides the sum of squared generated coordinate IQRs by the corresponding reference sum. Undefined coordinate ratios remain undefined rather than being stabilized by an arbitrary constant.

5.3 Content control and sensitivity analyses

The controlled reference panel has now produced all 210 expected feature vectors under three processing pipelines, without measurement failures or duplicate raw image hashes. Recomputing those vectors from retained source bytes reproduces all records. Both sensitivity scalers were fitted first, from 442 remeasurements of the same 221 historical development works. All 1,006 selected generated images produce their 3,018 expected vectors without measurement failures or duplicate raw hashes. Generated features are first accessed after terminal collection, and retained raw bytes support exact replay.

For painter p , let N_{pc} denote the number of reference works in content class c , N_p the total, and n_{pc} the number of available generated observations in a particular route/condition cell. The primary weights are

$$a_i = 1/N_p, \quad b_j = \frac{N_{pc}}{N_p n_{pc}} \quad \text{for } j \in c. \quad (2)$$

Each cell is assigned 72 slots: eight briefs per content class, each repeated three times. Missing outputs redistribute only their own class’s fixed mass; a required class with no measured output makes that comparison unavailable. The weighting is conditional on intended generated content, whose adherence has not been independently judged. Images are not discarded for failing to resemble a painter or for depicting the intended scene imperfectly. Effective weight size $1/\sum_j b_j^2$ is reported separately from the number of image files.

Primary energy discrepancy and the two spread ratios use all 31 standardized features. Color, spatial and texture subsets remain descriptive. Prespecified sensitivities use the collected generated mixture, one-third mass per content class, only unambiguous reference annotations, only retained reference files with native short sides at least 1024 pixels, and initial-attempt successes alone. The annotation and native-size restrictions redefine their reference class masses from the retained subset and are labeled accordingly. A 256-pixel short side and common JPEG encoding at quality 90 with 4:2:0 subsampling additionally probe processing sensitivity. No histogram, palette or texture matching is performed.

For the new disjoint original/original baseline, each of 999 deterministic draws selects two nonoverlapping sets of $\lfloor N_{pc}/2 \rfloor$ works per class and a generated set with the same class counts, without replacement. All sets retain the declared class masses. The resulting group sizes are 18 for Monet and 15 for Cézanne; the latter includes only one water work per group. The repeated draws characterize this finite panel and do not create independent reference samples.

Secondary coverage uses an original-centered ball with radius r_i equal to the distance to the third nearest distinct reference work:

$$\hat{C} = \sum_i a_i \mathbf{1} \left\{ \min_j \|x_i - y_j\|_2 \leq r_i \right\}. \quad (3)$$

We report all-available coverage and 100 uniform generated subsamples of size $\min(N_p, n_p)$, without replacement. These neighborhood measurements depend on sample size and representation and are not artistic-completeness scores.

Fixed linear/RBF kernel-ridge detection uses six whole-brief folds and disjoint reference works, with no tuning or PCA input. Reference folds are assigned by content-stratified identity hashes, so sparse classes are distributed across folds. Training assigns equal total mass to original and generated labels and matches content masses within each label; held-out balanced accuracy and AUC use the same target masses. Eight folds over assigned collection batches provide a timing sensitivity; the batches do not identify independent service sessions. Every split and prediction is retained. For visualization, each painter has one common PCA fit with half the mass on originals and half shared equally across seven generated cells, using the primary content weights. Original-only PCA is also retained. Every point is shown under shared within-painter

axes. Painter specificity compares each named cell with both reference painters under the same one-third-per-class mixture, separately from its absolute discrepancy.

5.4 Assignment, execution and missingness

The reference, prompt inventory, endpoints, scaling and analysis are committed before evaluation features are extracted. Repetitions are assigned to eight batches with originally planned offsets 0, 3, 6, 9, 24, 27, 30 and 33 hours. Within-route condition order is randomized, with at most one active call per route and globally staggered starts at least five seconds apart. All attempts, refusals, technical retries and actual timestamps are preserved.

Three interrupted collection components remain permanently closed: a generic HTTP 400 guard stopped on a moderation refusal; a later complete FLUX 502 stopped on missing cost; and the waiting recovery closed when the user requested immediate completion. Each correction has disjoint, predecessor-bound evidence and precedes all generated-feature measurement. The final 504 untouched slots run consecutively in batches 4–7 on 7 September, 00:29:19–02:01:37 UTC, under unchanged scientific assignments and rate limits. The complete 33-hour schedule was not executed. The final finite-slot view retains both refusals and the sole technical retry; it does not silently replace a slot or claim independent backend sessions.

Prospective synthetic qualification passes for the eight paired all-31 prompt contrasts: six named-versus-free and two detailed-versus-generic OAuth contrasts. Two fixed-seed phases each use 10,000 trials in 24 null cells covering pair counts, contribution heterogeneity, window drift and endpoint dependence. Actual tests use 99,999 Monte Carlo sign draws and Holm adjustment. Their sharp null concerns availability and features under the randomized slot policy, conditional on the finite reference, with no interference. Service carryover and retry-induced timing remain limitations. This qualification does not validate absolute-distribution p-values, oeuvre-sampling intervals or actual image-space power. Absolute gaps and leave-batch/brief ranges remain descriptive. Ordinary image-level bootstrap intervals are not justified by the number of files alone. No significance-based sampling extension, silent replacement of missing outputs or retrospective change to a sealed endpoint is allowed. Human construct validation remains unperformed. Absolute discrepancy and spread are descriptive; only the eight prespecified prompt contrasts carry conditional randomization p-values.

5.5 Paired statistic and conditional interpretation

For a matched before/after pair (A_i, B_i) with fixed class-adjusted weight b_i , define $C(v) = \sum_j a_j \|v - x_j\|_2$ and $T(v) = \sum_i b_i (\|v - A_i\|_2 + \|v - B_i\|_2)$. The contribution

$$c_i = b_i \{2[C(B_i) - C(A_i)] - [T(B_i) - T(A_i)]\} \quad (4)$$

satisfies $\sum_i c_i = \hat{\mathcal{E}}(X, B) - \hat{\mathcal{E}}(X, A)$. Thus the contrast retains the within-generated pairwise terms; it is not a contrast of mean-to-centroid distances. Complementary condition swaps produce $\sum_i s_i c_i$ for assignment signs $s_i \in \{-1, 1\}$. For the three-condition OAuth groups, each pairwise comparison conditions on the third condition’s position in the uniformly randomized order.

A contrast uses complete measured pairs sharing a brief, repetition and assigned batch. If fewer than 24 pairs or 12 distinct briefs survive, or a required content class is absent, its endpoint is unavailable and retains $p = 1$ in the fixed eight-test family. This is distinct from the all-available descriptive comparison. Two-sided Monte Carlo p-values use $(1 + e)/(1 + 99999)$, where e counts sign-draw statistics at least as extreme in absolute value, including numerical ties. Leave-one-batch and leave-one-brief estimates assess sensitivity without being labeled confidence intervals. The sharp assignment null covers both availability and measured features for the full fixed

slot policy, including its bounded technical retry rule; it is stronger than equality of expected distances. Treatment-dependent service duration and retry delays can affect later calls, so the randomization calculation does not remove the no-interference limitation.

6 Controlled results

6.1 Proximity and spread are distinct

All 14 primary cells and all eight inferential endpoints are available. Table 1 compares the full distributions, not just painter means. Every named condition has less total empirical variance than its reference (ratios .306–.756). Detailed named ratios are .354–.756; the two generic-named OAuth cells are .306 and .395. Naming also contracts spread relative to the corresponding artist-free condition in all six route/painter comparisons. However, paid artist-free ratios are 1.069–1.535. The data therefore support concentration of these painter-conditioned collections, not an intrinsic claim that generated images always have less variation.

Painter	Route	Prompt	Energy	Variance	IQR sum	RBF BA
Monet	NB2	Free	4.188	1.086	1.131	0.939
	NB2	Named	3.342	0.756	0.966	0.906
	FLUX	Free	1.990	1.071	0.941	0.901
	FLUX	Named	1.365	0.354	0.377	0.953
	OAuth	Free	2.567	0.863	1.176	0.960
	OAuth	Named	2.333	0.478	0.586	0.962
	OAuth	Generic	2.370	0.306	0.330	0.987
Cézanne	NB2	Free	4.704	1.535	1.371	0.970
	NB2	Named	2.886	0.716	0.785	0.977
	FLUX	Free	1.700	1.069	0.950	0.903
	FLUX	Named	0.804	0.492	0.567	0.893
	OAuth	Free	1.941	0.787	0.928	0.958
	OAuth	Named	1.884	0.512	0.461	0.920
	OAuth	Generic	2.166	0.395	0.299	0.918

Table 1: Primary 512-pixel, all-31, reference-content-weighted results. Variance and squared-IQR-sum values are generated/original ratios. BA holds out whole briefs and distinct original works. Each cell has 72 images except Cézanne/OAuth/generic (70); references have 38 Monet and 32 Cézanne works. NB2 denotes Nano Banana 2; FLUX denotes FLUX.2 Max; OAuth is the GPT Image 2 service. Named uses detailed briefs; Generic uses generic named briefs.

The contraction is uneven across feature families. Among the eight named cells, color variance ratios range .291–.745 and digital texture .143–.460. Spatial ratios range .665–1.520, with four detailed named cells above one. Thus, lower aggregate spread does not mean lower variation in every type of visual property. These family results are descriptive and do not identify a physical brushstroke or perceptual-diversity mechanism.

6.2 Scatter overlap coexists with full-space detectability

The balanced joint PCA retains 48.4% of weighted variation in its first two components for Monet and 46.0% for Cézanne. The clouds overlap in these views (Figure 2), with route- and condition-specific shifts and concentration. Original-only PCA, retained in the report, explains 55.6% and 52.2% in two dimensions and also shows overlap. Neither projection supports a claim of perfectly disjoint distributions.

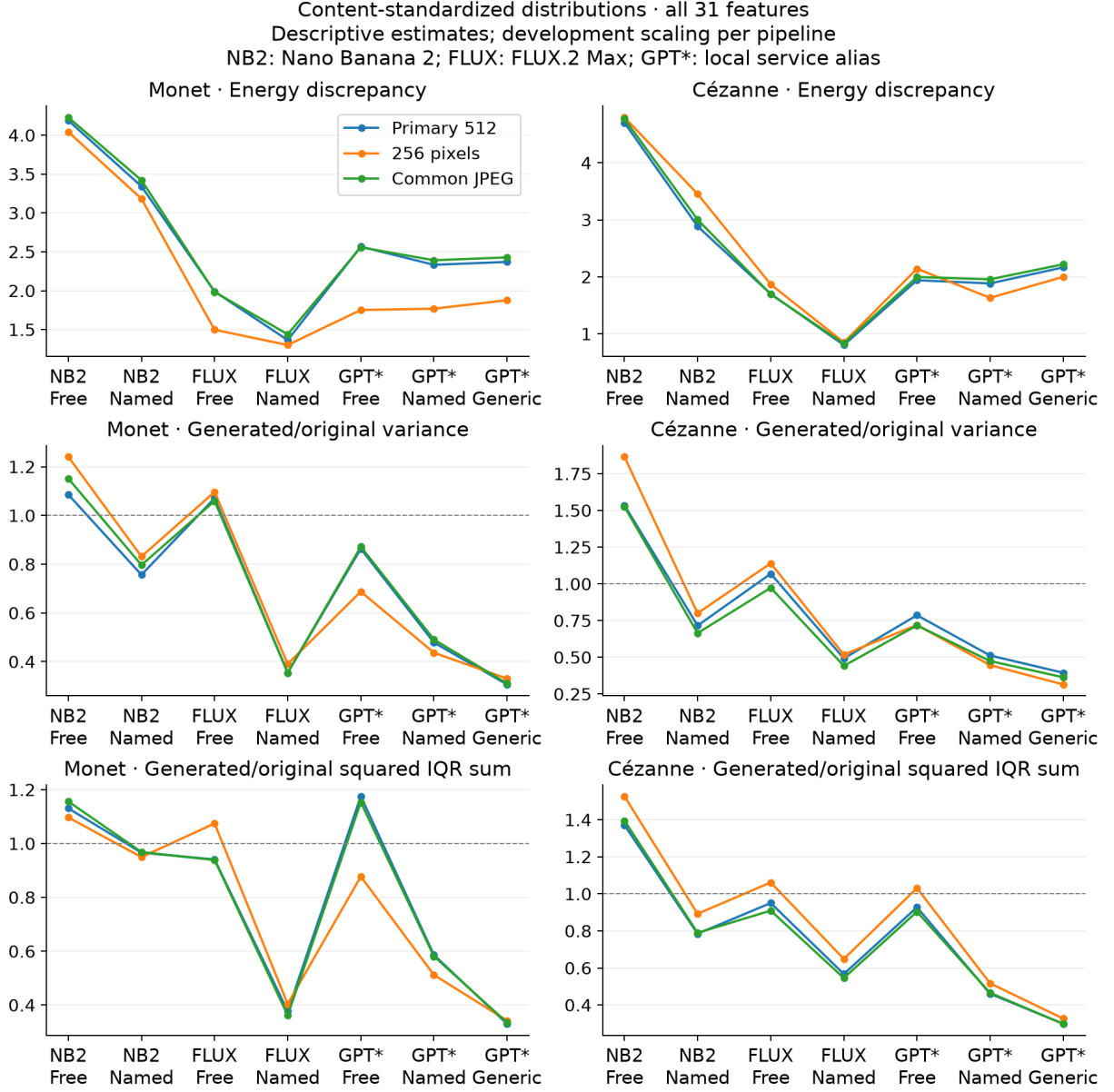


Figure 1: Full-distribution discrepancy and two spread summaries across the three fixed processing pipelines. Named means detailed named; Generic means generic named. Every pipeline uses its own development-only scaler. Dashed lines mark a spread ratio of one, not a calibrated equivalence boundary.

In all 31 features, whole-brief/work-disjoint RBF balanced accuracy is .893–.987 across all 14 cells, including artist-free controls. Linear accuracy is .837–.969. Holding out assigned batches gives .889–.987 (RBF) and .822–.969 (linear). These fixed-model descriptive scores establish practical detectability within the recorded evaluation splits. They do not identify which differences represent painter style, nor provide an absolute distribution-equality test.

6.3 Painter naming improves proximity on the paid routes

All six primary named-minus-free energy point estimates are negative (Table 2). The four paid-route contrasts reject the fixed conditional sharp null after Holm adjustment over eight tests. Their improvements range from .625 to 1.818 energy units. Every leave-one-brief and leave-one-batch estimate for those four contrasts remains negative. These ranges describe the finite design; they are not confidence intervals.

Neither OAuth naming contrast rejects after adjustment (Holm $p = .19308$ for Monet and $p = 1$ for Cézanne). The generic-to-detailed OAuth contrast is small for Monet ($-.0375$, Holm $p = 1$) and larger for Cézanne ($-.2723$, Holm $p = .0928$), using 70 complete Cézanne pairs because both refusals occur in its generic condition. Non-rejection does not establish no effect. Detailed named OAuth images have more empirical spread than generic ones for both painters, but this comparison changes content specificity as well as wording; it is not a pure length effect.

Route	Painter	Contrast	Pairs	Energy change	Raw p	Holm p
NB2	Monet	Name	72	-0.8462	0.00062	0.0031
NB2	Cézanne	Name	72	-1.8176	10^{-5}	8×10^{-5}
FLUX	Monet	Name	72	-0.6246	10^{-5}	8×10^{-5}
FLUX	Cézanne	Name	72	-0.8956	10^{-5}	8×10^{-5}
OAuth	Monet	Name	72	-0.2339	0.06436	0.19308
OAuth	Cézanne	Name	72	-0.0561	0.61624	1
OAuth	Monet	Detail	72	-0.0375	0.773	1
OAuth	Cézanne	Detail	70	-0.2723	0.0232	0.0928

Table 2: Prespecified two-sided paired tests, 99,999 Monte Carlo sign draws, Holm family size eight. Name compares artist-free with detailed named; Detail compares generic named with detailed named. Negative values mean closer after the intervention, on complete matched pairs. The sharp null includes availability and features under the fixed slot/retry policy and assumes no interference.

6.4 Finite-reference baselines, coverage and specificity

Across the 14 matched-sample comparisons, original/original energy medians range .697–.753, while original/generated medians range 1.170–5.232. Every generated comparison has a larger median than its corresponding original split. The lowest generated median is Cézanne/FLUX/named (1.170 versus .718); its subsample ranges overlap the original baseline. These finite-panel draws do not establish a population rejection or an equivalence threshold.

FLUX named outputs have the lowest primary energy among the tested routes for both painters (1.365 Monet, .804 Cézanne), together with the highest named all-available neighborhood coverage (.789 and .750). Equal-size subsample coverage medians are lower (.737 and .672). For Monet/FLUX, naming improves energy from 1.990 to 1.365 and coverage from .553 to .789, while variance contracts from 1.071 to .354. Proximity, spread and neighborhood coverage can therefore move in different ways; a single distance cannot describe all three.

Under equal content-class masses, five of six detailed named cells are closer to their own reference painter than to the other painter. Nano Banana 2’s Monet condition is the exception:

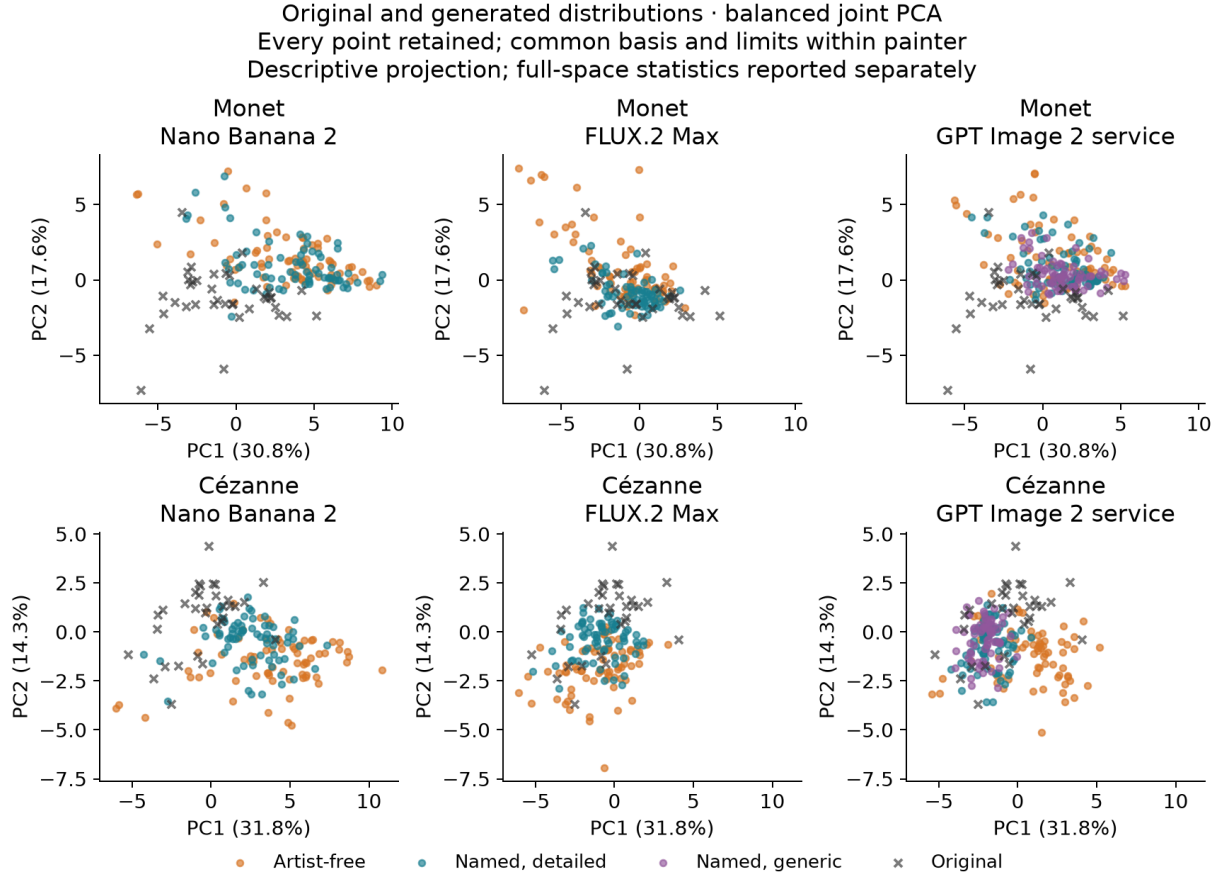


Figure 2: Common painter-specific PCA with every original and generated point retained. Each painter uses one basis and shared limits across routes. Originals receive half the fit weight; generated cells share the other half. Overlap in two projected dimensions coexists with strong held-out full-space detection.

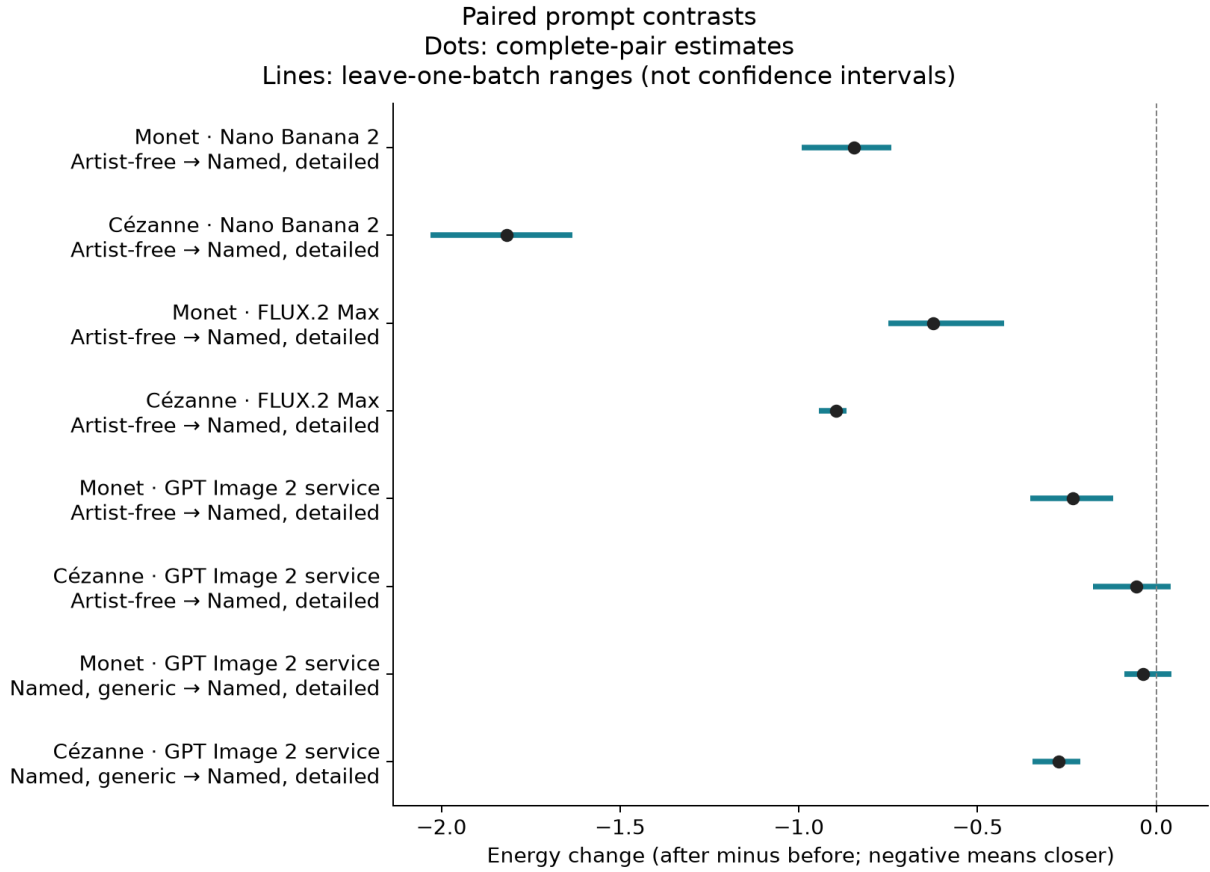


Figure 3: Complete-pair estimates and leave-one-assigned-batch ranges. Shared service state and timing dependence remain limitations of the conditional tests; the displayed lines are not uncertainty intervals for a painter’s oeuvre.

energy 3.687 to Monet versus 3.488 to Cézanne. Improved proximity to a requested target thus does not guarantee painter specificity. This two-painter comparison is descriptive and is not a validated recognition task or a general model leaderboard.

6.5 Processing and reference sensitivities

For the eight named cells, total-variance ratios remain below one under both fixed processing sensitivities: .315–.831 at 256 pixels and .310–.797 after common JPEG encoding. All four paid-route naming effects retain their negative descriptive energy direction under both. OAuth is less stable: Monet’s naming difference changes from $-.234$ in the primary pipeline to $+.017$ at 256 pixels. No additional significance family is introduced for these sensitivities.

Collected-mixture, equal-content and unambiguous-reference analyses retain aggregate variance ratios below one for all eight named cells. Some IQR-based ratios exceed one after reweighting or reference restriction. The native-1024 restriction is particularly limited: only four Monet and 19 Cézanne reference files remain. Nano Banana 2’s Monet variance ratio becomes 1.107 in that primary subset (1.123 at 256 pixels; 1.178 after JPEG). This is a material exception to uniform robustness and a change in the finite comparison target. It cannot serve as a strong independent capture control. Initial-only analysis preserves all named cells; its only additional exclusion is the retried FLUX artist-free slot.

7 Discussion

The controlled results support the narrow hypothesis that these painter-named collections differ from the recorded original-painting distributions and occupy less aggregate feature variation. They also revise an overly broad version of that hypothesis: the paid artist-free controls can be more dispersed than the originals, spatial features need not contract, and a small reference restriction reverses one aggregate result. The evidence is about the tested conditional collections and finite digital reference panels.

The strongest prompt finding is improved proximity with contracted spread on both paid routes. This is consistent with painter naming selecting a narrower region of the model’s output distribution, but the experiment does not identify that internal mechanism. Better energy fit can coexist with higher coverage, strong detection and incomplete painter specificity. Calling all of these outcomes “style fidelity” would conceal useful distinctions.

The next research priority is construct and reference validation, rather than another large generation sweep. Blinded human judgments of painter recognizability and within-set variation could test whether the interpretable measurements track perceived style. Independent content annotation and larger, capture-documented reference panels would address the weakest controls. Learned features can later provide a complementary representation, with their own processing checks and human calibration. More generated files alone would not resolve these limitations.

8 Limitations and reproducibility

The original corpus is a finite collection of available digital surrogates, not a probability sample of each painter’s oeuvre. The new reference panel is small and availability constrained. Service aliases and live model slugs lack attested immutable weight snapshots. Prompt repetition, shared briefs and service state can induce dependence. Title-based development labels and later maintainer-LLM visual annotations are not independent expert judgments. All review so far is maintainer/LLM self-review; none is institutionally independent.

The paper source, compact manifests, exact prompts, preprocessing code, report tables, split

memberships and hashes are retained in the repository. Raw artwork and complete generated-image responses remain in an ignored local evidence workspace; a Git checkout alone does not reconstruct those bytes. Acquisition licenses and source identities are recorded, and failed responses remain evidence. The new study uses its own namespace and does not modify historical terminal runs.

Human evaluation is a prepared follow-up, not an executed component. Expert review could assess painter recognizability and perceived variation using blinded, balanced stimuli and independent annotation. Learned representations are also deferred; adding a learned evaluator would not automatically establish that its distance is calibrated to human style judgments. Neither a collaborator’s participation nor authorship is presumed in this prototype.

A Reproduction entry points

From the repository root, replay the completed controlled numerical analysis and all report files without provider calls:

```
uv run --locked --extra analysis python -m
latent_art_bench.painter_distribution_study_v1.analysis_publication check

uv run --locked --extra analysis python -m
latent_art_bench.painter_distribution_study_v1.main_report check
```

A separate `immediate_results generated --check` reopens retained raw bytes and recomputes all 3,018 feature vectors. The original calculation reached its JSON exporter, which rejected four NumPy Boolean rejection flags before writing any result file. The disclosed publication adapter converts only those flags to native Booleans and requires exact value equality after JSON round-trip. It uses a separate publication namespace and leaves the frozen statistical calculation unchanged. Its source, inputs and publication freeze are committed before writing the completed analysis. This representational correction is not a scientific endpoint amendment.

Historical diagnostics replay through the same package’s `diagnostics check` entry point. Exact dependencies are pinned by `uv.lock`; no learned encoder is run. Current operational status and commands are in `docs/STATUS.md` and `docs/DISTRIBUTION_STUDY_WORKFLOW.md`. Raw image bytes require the separately retained local evidence workspace. No completed collector or writing stage should be restarted to reproduce a report.

References

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